ARSITHA SATHU

Edison, NJ | (202) 222-5321 | sathuarsitha@gmail.com | https://www.linkedin.com/in/arsitha-reddy

SUMMARY

AI Engineer and software developer building end-to-end generative AI applications, including evidence-grounded document intelligence, retrieval-augmented conversational assistants, and multi-agent learning workflows. Hands-on experience with Python, FastAPI, OpenAI/Azure OpenAI, vector search, multimodal document processing, LLM verification, and cloud deployment. Combines applied NLP experience with reliable backend engineering and human-in-the-loop review systems.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript, TypeScript
Generative AI & LLMs: OpenAI API, Azure OpenAI, GPT-4o, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Orchestration, AI Agents, Structured Outputs, Function Calling
AI/ML & NLP: PyTorch, Hugging Face Transformers, Embeddings, Semantic Search, Vector Databases, LLM Evaluation, Multimodal AI, Pandas, NumPy
Frameworks & Backend: Fast API, REST APIs, Spring Boot, Node.js, Microservices, React.js
Databases: PostgreSQL, MySQL, ChromaDB, AWS Aurora PostgreSQL
Cloud & DevOps: Azure AI Services, Azure Container Apps, Azure Key Vault, AWS (EC2, S3, Lambda), Docker, Kubernetes, Git, GitHub, CI/CD
Tools & Testing: Postman, Swagger/OpenAPI, Jenkins, JUnit, SonarQube

PROFESSIONAL EXPERIENCE

Knowia Tech | Software Engineer *April 2026 – Present*

- Developed an end-to-end Generative AI platform using Python and LLMs to transform lengthy government reports and PDFs into structured, interactive web stories with source-grounded summaries and visualizations.
- Engineered multi-stage document processing pipelines using Python and AI-powered extraction techniques to extract text, tables, figures, and numerical data from complex PDFs while preserving source references for factual accuracy.
- Implemented multimodal AI workflows combining vision models and chart analysis to interpret figures, extract quantitative insights, and generate interactive data visualizations.
- Designed LLM-based content generation and evaluation workflows with structured prompts, source validation, numerical consistency checks, and automated detection of unsupported claims.
- Developed human-in-the-loop AI editing workflows that support content refinement, figure replacement, version history, and revalidation of AI-generated content before publication.

Blue Horse Digital | Software Engineer *Jun 2025 – April 2026*

- Developed a conversational AI assistant for U.S. farmers using Python, Azure OpenAI (GPT-4o), and Retrieval-Augmented Generation (RAG) to provide context-aware answers to USDA policy and agricultural program questions.
- Implemented document ingestion, text chunking, embeddings, and vector-based semantic search to retrieve relevant policy information and ground LLM responses in trusted sources.
- Integrated Azure Speech Services for speech-to-text and text-to-speech capabilities, enabling voice-based interaction with the AI assistant.
- Built an automated PDF form-filling workflow using Python and PyPDF to map conversational user responses to form fields and generate completed documents.
- Containerized and deployed AI backend services using Docker and Azure Container Apps, securing application credentials through Azure Key Vault.

Infosys Limited | Software Engineer *Oct 2021 – Jul 2023*

- Developed Java/Spring Boot microservices and REST APIs for healthcare compliance workflows, including validation, record processing, search, reporting, and PostgreSQL integrations.
- Improved API response times by 25–30% through SQL query optimization and indexing, and maintained 90%+ JUnit test coverage.
- Diagnosed production issues across services and data layers; supported CI/CD, automated quality checks, and releases using Jenkins, Git, and SonarQube.

PROJECTS

Clip Learn | Multi-Agent AI Learning Assistant *Python, FastAPI, GPT-4o, ChromaDB*

- Orchestrated five specialized agents for transcript extraction, content generation, semantic search, translation, and lecture quality review, streamed progress with server-sent events.
- Built transcript-grounded Q&A with timestamp references, auto-generated summaries and flashcards, and on-demand support for 10 languages.

Abstractive Text Summarization | Transformer Fine-Tuning *Python, PyTorch, Hugging Face, mT5*

- Fine-tuned mT5 for Telugu abstractive summarization on 200+ narratives, created preprocessing and evaluation workflows using ROUGE and human review.

Threat Ledger | AI-Assisted Security Analytics *OpenAI API, PostgreSQL, Next.js*

- Normalized 200K+ security events, correlated 200+ alerts into fewer than 20 attack campaigns, and generated AI-powered threat summaries from supporting event data.

EDUCATION

M.S. in Computer Science | George Mason University | May 2025

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (May 2026)
AWS Certified Cloud Practitioner (Feb 2026)